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Sub : Chemistry Assignment 7

Q1

What is isomerism in organic chemistry?
Isomerism in organic chemistry is a phenomenon shown by two or more organic compound having the same molecular formula but different properties due to difference in arrangement of atoms along the carbon skeleton (structure isomerism) or in space (stereoisomerism).

Q2

What is there and erythro?
Erythro and threo are two configurations in which molecules are written when the molecules have a chiral carbon atom. Erythro is the configuration when the same groups are on the same side of the carbon atom and threo is the configuration when the same groups are on the opposite side of the carbon atom.

Q3

What is diastereoisomerism?
Diastereomers are defined as non-mirror image non-identical stereoisomers. Hence they occur when two or more stereoisomers of a compound have different configurations at one or more of the chiral stereocenters and are not mirror images of each other.

Q4

What are the types of isomerism?
Two main forms of isomerism are structural or constitutional isomerism, in which bond between the atoms differ, and stereoisomerism or spatial isomerism, in which the bond are the same but the relative positions of the atoms differ.

Q5

What are functional isomers?
Functional isomers are structural isomers that have the same molecular formula, but the atoms of the same element, but the atoms are connected differently ways. E.g. that the grouping are different. These groups of atoms are called functional groups, functional groups.

Q6

What are isomers? Give their classification with 4 examples.

Ans. Isomerism: Isomers are compounds that have the same molecular formula but different arrangement of atoms, resulting in different arrangement of atoms resulting in different structure of properties.

* Classification of Isomerism

classification of isomerism:

Isomerism is mainly classified into two types:

1. structural isomerism (constitutional isomerism)
- structural isomers have the same molecular formula but different connectivity/arrangement of atoms.

It is further classified into:

1. chain (skeletal) isomerism — difference in the carbon skeleton (branching)

Example: C_5H_{12}

n-pentane and 2-methylbutane.

2. position isomerism — Same carbon skeleton and functional groups, but the functional groups / substituent is at different positions.

Example: C_3H_7Cl

3. chloropropane and 2-chloropropane.

3. functional group isomerism — Same molecular formula but different functional groups.

Example: C_3H_6O

propanal (aldehyde) and propanone (ketone)

Metamerism — different alkyl groups are present on either side of a polyvalent

divalent functional groups such as $-O-$ or $-S-$.

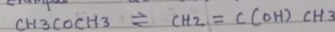
Ex:

$C_4H_{10}O$: ethoxyethane ($C_2H_5-O-C_2H_5$) and 1-methoxypropane ($CH_3-O-C_3H_7$)

5.

Tautomerism — Isomers that differ in the positions of a proton and a double bond and exist in equilibrium.

Example: Keto-enol tautomerism



6.

Ring-chain isomerism — One isomer has an open-chain structure while the other has a ring structure.

Ex:

C_3H_6

propene ($CH_3-CH=CH_2$) and cyclopropane.

2.

Stereoisomerism —

Stereoisomers have the same molecular formula and same connectivity of atoms but differ in their three-dimensional arrangement.

It is mainly classified into:

- * Geometrical isomerism — cis-trans / E-Z isomerism

Ex: cis-2-butene and trans-2-butene

- * optical isomerism: isomers that are non-superimposable mirror image of each other. Ex: (R) and (S) lactic acid.